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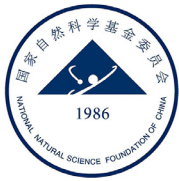
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Article

Thin, soft, 3D printing enabled crosstalk minimized triboelectric nanogenerator arrays for tactile sensing



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ABSTRACT

With the requirements of self-powering sensors in flexible electronics, wearable triboelectric nanogenerators (TEGs) have attracted great attention due to their advantages of excellent electrical outputs and low-cost processing routes. The crosstalk effect between adjacent sensing units in TEGs significantly limits the pixel density of sensor arrays. Here, we present a skin-integrated, flexible TENG sensor array with 100 sensing units in an overall size of 7.5 cm × 7.5 cm that can be processed in a simple, low-cost, and scalable way enabled by 3D printing. All the sensing units show good sensitivity of 0.11 V/kPa with a wide range of pressure detection from 10 to 65 kPa, which allows to accurately distinguish various tactile formats from gentle touching (as low as 2 kPa) to hard pressing. The 3D printing patterned substrate allows to cast triboelectric layers of polydimethylsiloxane in an independent sensing manner for each unit, which greatly suppresses the cross talk arising from adjacent sensing units, where the maximum crosstalk output is only 10.8%. The excellent uniformity and reproducibility of the sensor array offer precise pressure mapping for complicated pattern loadings, which demonstrates its potential in tactile sensing and human-machine interfaces.

1. Introduction

Recently, flexible and wearable electronics have attracted extensive attention due to their promising potential in the internet of things [1–6] and human-machine interfaces [7–10]. Tactile sensing converts human activities and motions related to touch information into electrical signals, exhibiting significant roles in wearable electronic devices [9,11,12]. Tremendous efforts in flexible and wearable tactile sensors have been made based on various sensing mechanisms, such as piezoresistive [13,14], piezoelectric [15–18], and capacitive [19,20] based ones. Equally important in wearable electronics, power modules playing significant roles, however, are still limited by their size, flexibility, and applicability [21–26]. As a result, self-powered electronics as an emerging technology has been considered as the candidate for next generation wearable sensing systems. To date, self-powered electronics have been widely studied based on different technologies, including piezoelectric nanogenerators (PENGs) [27,28], triboelectric nanogenerators (TEGs) [29–33], and thermoelectric generators (TEGs) [34,35].

Among them, TENGs, which convert mechanical energy into electricity via the relative movement of a pair of coupling electrodes, have been proved to be excellent candidates for self-powered tactile sensing devices due to the inherent merits of low cost, simple fabrication process, and broad material availability with excellent electric performance [30,36–38].

In the material selection of TENGs, it is known that materials with opposite electron affinity can serve as the coupling pairs of triboelectric layers [39]. Among diverse functional materials for triboelectric layers, polydimethylsiloxane (PDMS) is an ideal candidate due to its excellent flexibility, simple fabrication process, and good electronegativity [9,32,40,41]. While the electrodes of TENGs typically adopt metallic foils or thin film, which may limit the stretchability. In stretchable electronic technologies, metallic traces with predesigned patterns are often used to increase stretchability [23,42–45]. Considering the complicated microfabrication processes and unconventional integration routes of stretchable metallic traces, nanomaterials, i.e. graphene [46,47], carbon nanotubes [48,49], metal-based materials, silver nanowires (Ag

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NWs) [50,51] and gold nanowires (Au NWs) [11,52] have been reported for flexible TENGs. Among them, Ag NWs exhibit excellent mechanical performance, robust durability, high electrical conductivity, and high throughput deposition routes such as printing like spray-pyrolysis, and thus offer the possibility of developing low-cost and scalable TENGs [53].

TENGs have great potential for self-powered sensors, while the crosstalk effect caused by the interaction between adjacent sensing units and electrodes limits the pixel density and practical applications. Several methods have been adopted to weaken the crosstalk effect, such as introducing a shielding/screening layer or creating a height difference between the electrode and the sensors. For instance, Yao et al. [29] and He et al. [54] introduced Ag NWs as shielding layers to suppress the crosstalk effect. Wang et al. [55] introduced a height difference of 4.4 mm between the sensing area and the interconnected electrodes to suppress the crosstalk effect. However, introducing additional shielding layers or using a height difference strategy will not only complicate the fabrication process but also compromise the flexibility due to the mechanical mismatch of different functional layers [55]. Besides, introducing an extra shielding layer will also decrease the optical transmittance of the device. Hence, it is of great significance to design a skin-integrated, flexible TENG array with simple structure, high optical transmittance, and the capability of reducing crosstalk.

Here, we report the materials, devices, processing methods, and integration strategies for soft, transparent, self-powered single electrode TENG sensor arrays, based on 3D printed soft substrate, solution-processed Ag NWs electrode, and PDMS triboelectric layers, which offers capabilities of cross talk free tactile sensing. The sensor array can be fabricated via an easy and low-cost way and enable suppresses the crosstalk effect between adjacent sensing units at the same time. Specifically, Ag NWs are sprayed onto a 3D printed soft transparent substrate to serve as the transparent electrodes, and PDMS is cast on top of the electrodes as the triboelectric and sealed layer. The 3D printed pre-patterned substrate allows the triboelectric layer to independent sense forces in each unit, which can greatly suppress the crosstalk effect due to different dielectric properties between PMDS and the substrate. The electrical performance of the sensor array, including sensitivity, responses to different external stimuli and output performance after over 2000 loading cycles and 100 bending cycles were studied to characterize its electrical stability and mechanical durability. Moreover, the crosstalk outputs generated by adjacent sensing units at three types of loading positions were studied. Finally, demonstrations of loading forces in various patterns on a 10×10 soft sensor array proved the accuracy and crosstalk free behaviours of the system.

2. Experimental section

2.1. Fabrication process of the sensor array

Firstly, the soft, transparent substrate with electrode patterns was fabricated by a commercial digital laser resin printer (HALOT-SKY, CREALITY) with polyurethane acrylate. Especially, the 3D model of the substrate was designed in 3D software SOLIDWORKS (Version 2019, DASSAULT SYSTEMS). Subsequently, the model was converted into STL format before importing into the pre-processing software HALOT BOX (Version 2.0, CREALITY) to get a sliced document. The thickness of each slice was set to $50 \mu\text{m}$ and the laser exposure time of each slice was set to 3 s. It is noted that the exposure time of the first 5 layers was set to 30 s to get a strong bond between the soft substrate and the building platform. After finishing all layers, the substrate was post-processing with ethanol followed by UV exposure for 2 min to remove and cure the residual liquid polymer on the surface. Secondly, the soft substrate was exposed to UV-induced ozone for 5 min to increase the chemical group on the surface, which can greatly increase the bonding strength between Ag NWs electrode and the substrate and reduce the agglomeration of Ag NWs. Then, Ag NWs were spray-coated on the 3D printed substrate at a

discharging speed of $1 \mu\text{L/s}$ with the assistance of a thin steel mask to form the patterns of the electrodes. Finally, PDMS (with the cross-linker ratio at 10:1) was cast on the top of the substrate followed by a scraping process with a steel ruler. Then the substrate covered with Ag NWs electrodes and PDMS was put into a bake at 80°C for 2 h to cure the top layer PDMS before vacuum pumping for 20 min to remove the bubbles. It is noted that the connection parts of the electrodes were sealed during the casting process of PDMS for further connection to external circuits.

2.2. Characterization and performance analysis

The surface topographies of spray-coated Ag NWs electrodes were characterized by a scanning electron microscope (SEM, FEI Quanta 450) with 3D printed soft polymer serving as the substrate. The transparency of the sensor array was measured using Ultraviolet Visible Spectrometer (UV-VIS, Perkin Elmer Lambda 35) with air serving as the background. Besides, a resistance tester (Hepass HPS 2535) was utilized for testing the sheet resistance of Ag NWs electrodes. Furthermore, the open-circuit voltage, short-circuit current and the crosstalk of adjacent sensing units under three different loading points were characterized. Specifically, the open-circuit (OC) voltage of a single sensing unit was measured by a DAQ 6510 (Tektronix, US) data acquisition system and the sampling frequency was set to 60 kHz. The short-circuit (SC) current was calculated based on the relationship between voltage and resistance, where a resistor with resistance at $1 \text{ k}\Omega$ was connected in series to sensing units. The voltage was measured with PL 3516 (AD Instrument, New Zealand) with the sampling rate at 10 kHz to get a reasonable signal-to-noise ratio. The crosstalk voltages under different loading points were also measured by PL 3516 at 10 kHz with a low-pass Gaussian filter at 50 Hz to remove the noise signals.

3. Results and discussion

The flexible TENG sensing system can be processed layer by layer via a low-cost and scalable method. Fig. 1a, b show the schematic illustrations of the arrayed TENG sensing system, including a soft substrate, Ag NWs electrodes, and a PDMS sealing layer. As shown in Fig. 1e, the fabrication process begins with the soft substrate, where a soft and transparent substrate with patterned channels (Fig. 1d) was created by 3D printing commercially available polyurethane acrylate. The printed material owns the merits of robust durability, excellent flexibility, and satisfied optical transmittance. A thin layer of Ag NWs spray-coated on the substrate with a pre-designed pattern serves as the flexible and transparent electrodes (Fig. 1c). Finally, a thin layer of PDMS with a thickness of 0.2 mm cast on the top of electrodes acts as the triboelectric layer for the sensing units and the encapsulation layer. PDMS exhibits excellent flexibility and good electronegativity, and thus offers a great electrical performance of the device. The PDMS together with human skin or any other materials showing a positive electron affinity form a pair of electrodes to generate electric output when contact and separation occur between the interface, enabling TENGs to work in a single electrode mode (Fig. S1). Here, a 10×10 TENG sensor array with 100 sensing units and an overall dimension of $7.5 \text{ cm} \times 7.5 \text{ cm}$ in area, 0.4 mm in thickness was integrated. Each sensing unit took $4 \text{ mm} \times 4 \text{ mm}$ in a square area, with spacing 4 mm away from each other. The soft substrate combined with ultra-thin Ag NWs, and PDMS endows the sensor array with excellent flexibility under various kinds of mechanical deformations, such as bending (Fig. 1f), twisting (Fig. 1g), and rolling (Fig. 1h).

Transparency and conductivity are two key parameters of Ag NWs electrodes, which could greatly affect the optical transmittance and electrical conductivity of the device. It is well-known that a long deposition duration typically results in a thicker layer of Ag NWs with high electrical conductivity but low transparency. Fig. S2 presents the sheet resistance and optical transmittance of the Ag NWs thin film under various spray-coating parameters, where transparency over 60% can be

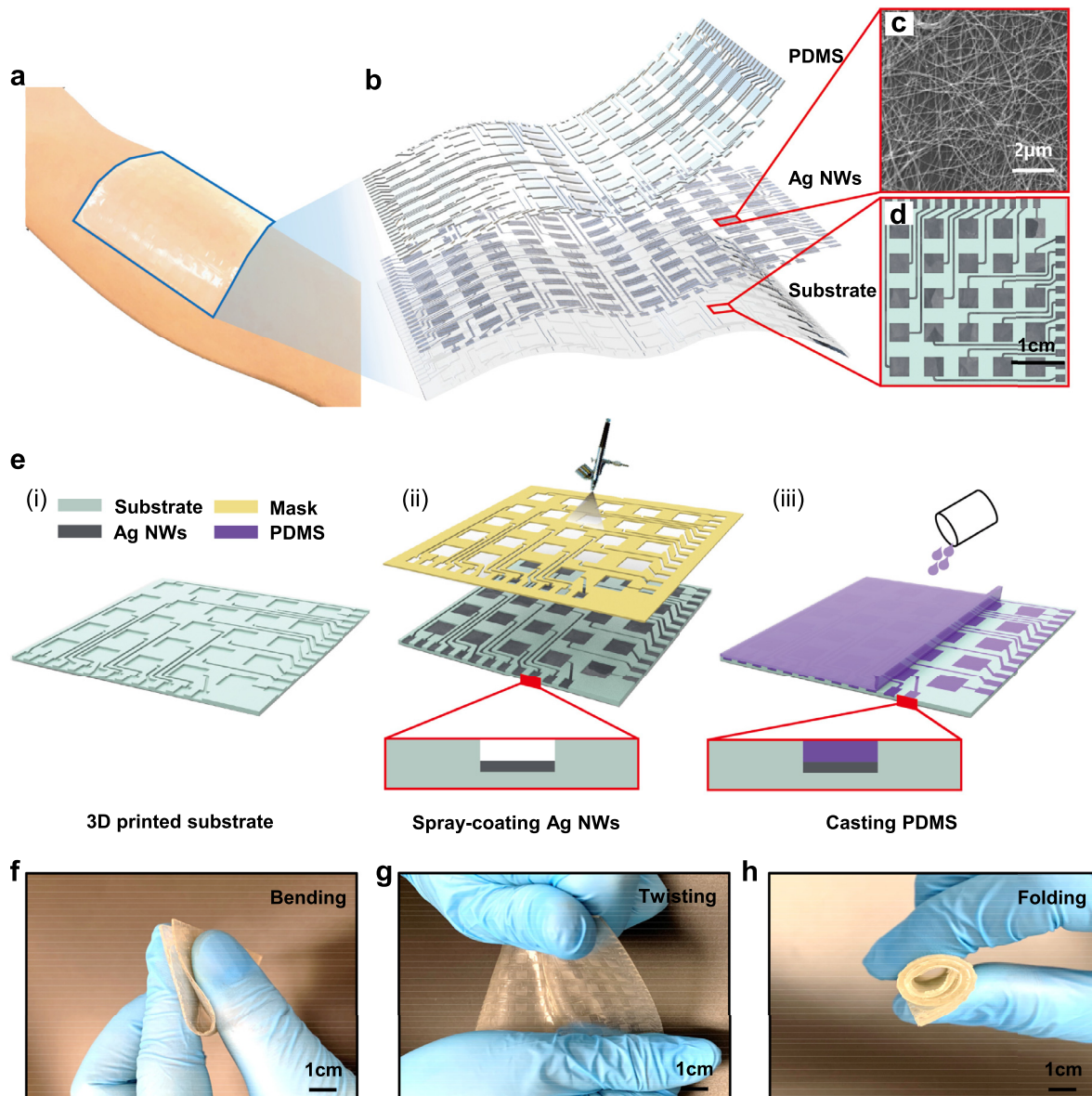


Fig. 1. Design and fabrication process of the sensor array. (a) Optical image of the e-skin mounted on the forearm of the volunteer. (b) Schematic illustration of the e-skin. (c) Scanning electron microscope (SEM) image of Ag NWs on the soft substrate. (d) Enlarged view of substrate with 5×5 pixels. (e) Schematic illustration of the fabrication process of the e-skin. (f-h) Optical images of the e-skin under different mechanical deformations, including bending, twisting and folding.

achieved after spraying 30 s (sheet resistance of $64 \Omega/\text{sq}$), while further increasing the duration to 50 s lowers the transparency to around 40% but also greater sheet resistance to $15 \Omega/\text{sq}$. Interestingly, the transparency of the device decreases dramatically at the wavelength of around 400 nm due to the absorption of the photocurable substrate. Here, a spray-coating duration of 40 s offers the optimized performance with balanced properties in both transparency (over 50%) and conductivity ($33 \Omega/\text{sq}$).

The electrical performance of each sensing unit determines the output performance of the TENGs array. Fig. 2a shows the study of electrical performance for a single TENG sensing unit, where different pressure loads ranging from 10 to 65 kPa were applied to the sensing unit at a frequency of 2 Hz. The open-circuit (OC) voltages generated by the sensing units under different loads were shown in Fig. 2a, where the maximum and minimum output OC voltage of 6.2 V and 0.79 V were associated with the loading pressure of 62.5 kPa and 13 kPa, respectively. Meanwhile, the OC voltage shows a positive linear correlation with the increase of the loading pressure [36,39]. The sensitivity of a

$4 \text{ mm} \times 4 \text{ mm}$ sensing unit can be calculated from the fitted line in Fig. 2a and turned to be 0.11 V/kPa , which offers a good sensitivity with a broad range of pressure responses and thus can be used in various sensing application scenarios. Fig. 2b presents the OC voltage outputs in a sensing unit as a function of time under various pressures, where consistent voltages were generated under the same loading pressure and differences were also very obvious when applied with different external loads. Meanwhile, the data of the sensor exhibits excellent time response with a great signal-noise ratio. Furthermore, the short-circuit (SC) current response to different external loads is presented in Fig. 2d, where the maximum SC current of $20.5 \mu\text{A}$ is associated with the maximum pressure of 62.5 kPa. Similarly, the SC current as a function of time under 6 types of external loads is presented in Fig. 2e, confirming the rapid time response and great stability of the sensing unit.

To further study the stability and durability of the soft TENG based sensing units, repeat loading and unloading pressure of 50 kPa at 10 Hz for multi cycles (>2000) was performed on a sensing unit (Fig. 2c). The output OC voltages were maintained at a similar value during multiple

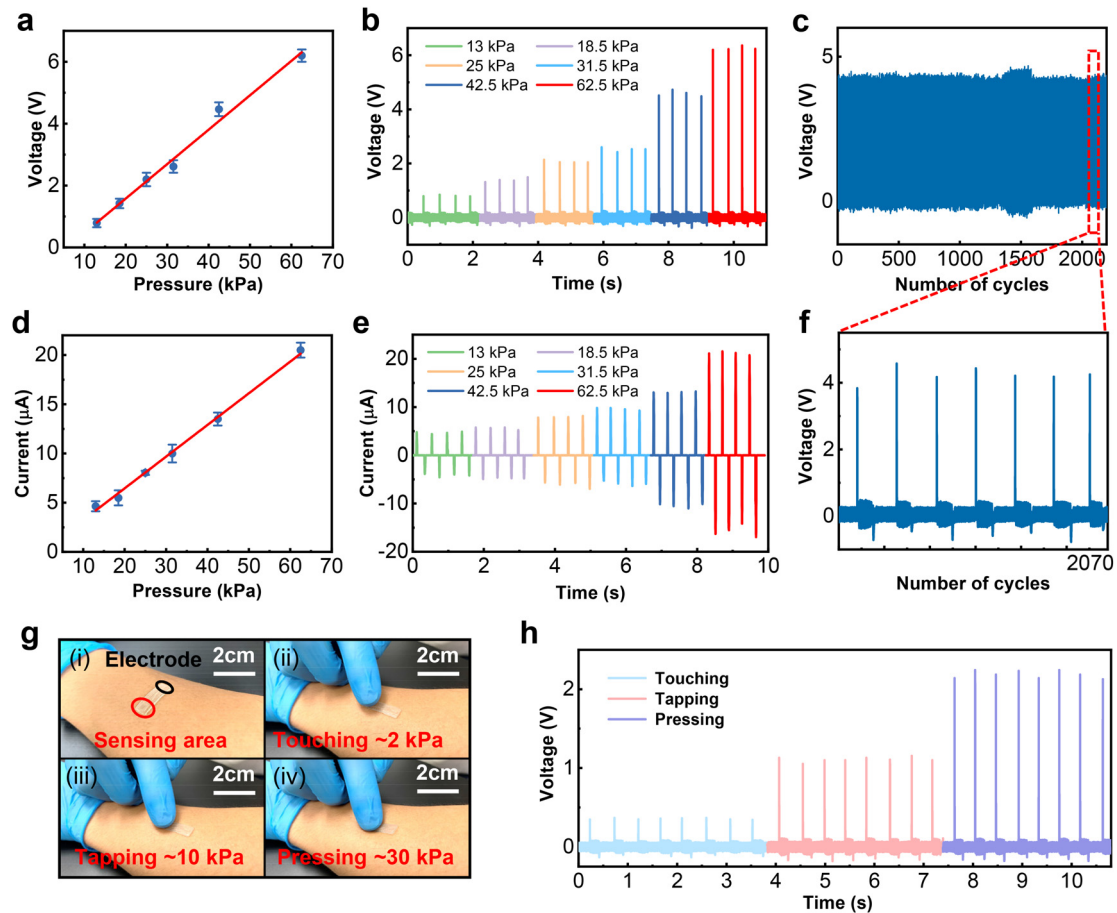


Fig. 2. Electrical characteristics of a single sensing unit. (a) The open-circuit voltages of a sensing unit under different external loads and the corresponding voltages varying with time (b) under a loading frequency of 2 Hz. (c) The output performance of a sensing units under 2100 loading cycles and the enlarged view of output voltage after more than 2000 loading cycles (f,d) The short-circuit current of a sensing unit under the corresponding loads in (a) and the current varying with time (e,g) Optical images of a sensing unit mounted on forearm of the volunteer (i) and under three types of loads, touching (ii), tapping (iii) and pressuring (iv). (h) The corresponding output voltages under the loads in (g).

cycles measurement, where the output voltage remained its initial value (Fig. 2f) after over 2000 loading cycles, indicating excellent stability and durability of the device. Furthermore, the electrical output performance of the sensing units wouldn't be affected even under a bending state (Fig. S3). Fig. S3 shows the OC voltage outputs of a sensing unit versus various loading pressures under two bending radii at 2 cm and 5 cm, where it can be seen the output voltages exhibited great stability regardless of bending radius. Fig. S4 presents the electrical output of the sensing unit under a pressure of 35 kPa at 4 Hz for over 100 cycles of bending deformation. No obvious voltage output changes were observed after bending hundreds of times, which proves the excellent flexibility and robust features of the sensing unit. Fig. 2g shows the sensing unit mounted on the forearm of a volunteer, which shows the possibility of acting as skin-integrated sensors to detect body activities and tactile information. During the test, three types of loads, including touching (2 kPa), tapping (10 kPa), and pressing (30 kPa) were applied (Fig. 2g(ii-iv)). The corresponding output voltages of the TENG sensor shown in Fig. 2g are presented in Fig. 2h, where the average OC voltage of the sensor of 0.34 V, 1.11 V, and 2.13 V associated with touching, tapping, and pressuring, respectively. The results prove the sensing unit can distinguish different tactile forces accurately from gentle touching (as low as 2 kPa) to hard pressuring.

Owing to the working mechanism of TENGs, electrostatic charge will be generated in adjacent sensing units and electrodes when external stimuli are applied on a sensing unit, which is known as the crosstalk effect. Crosstalk impedes the applications of TENG sensor arrays in precise

pressure mapping. In this work, a 3D printed substrate was developed with a pre-designed pattern, allowing to form the triboelectric layer PDMS independently for each sensing unit, which can greatly suppress the crosstalk caused by the interaction between adjacent sensing units and electrodes. To demonstrate the results of the suppressed crosstalk, three typical loading positions (Fig. 3a), including the sensing unit, the interval between two sensing units, and the intersection of four sensing units were loaded with an aluminum bar with the pressure of 30 kPa at a frequency of 3 Hz (Fig. 3b). Fig. 3c-e presents the average OC voltage output generated by 25 sensing units in a 5 × 5 sensor array. Fig. 3f-h shows the corresponding voltage as a function of time, which has been categorized based on the position of sensing units, and the specific output signals of each sensing unit can be found in Fig. S5. When the pressure was loaded on the sensing units (position 1 in Fig. 3a), an average output voltage of 3.9 V was generated, as shown in Fig. 3c, f. At the same time, very low output voltages, in the range of 0.32 to 0.51 V were generated in the 8 adjacent sensing units (Figs. 3c, f, and S5d). While the adjacent sensing signals are only 3.2–8.4% of the targeted output voltage (3.9 V), indicating excellent crosstalk suppression. The signals in the rest of 16 sensing units, ranging from 0.15 to 0.2 V can be considered as noise signals (Figs. 3f, and S5 d). When the loading position came to the interval of two sensing units (Fig. 3b-ii), small OC voltage output was generated in two adjacent units of 0.6 V and 0.53 V (Fig. 3d), which are 10.8% and 8.9% compared to that loaded on the sensing unit (3.9 V). The slight output can be attributed to two reasons, one is the incomplete contact between the loading stick and the sens-

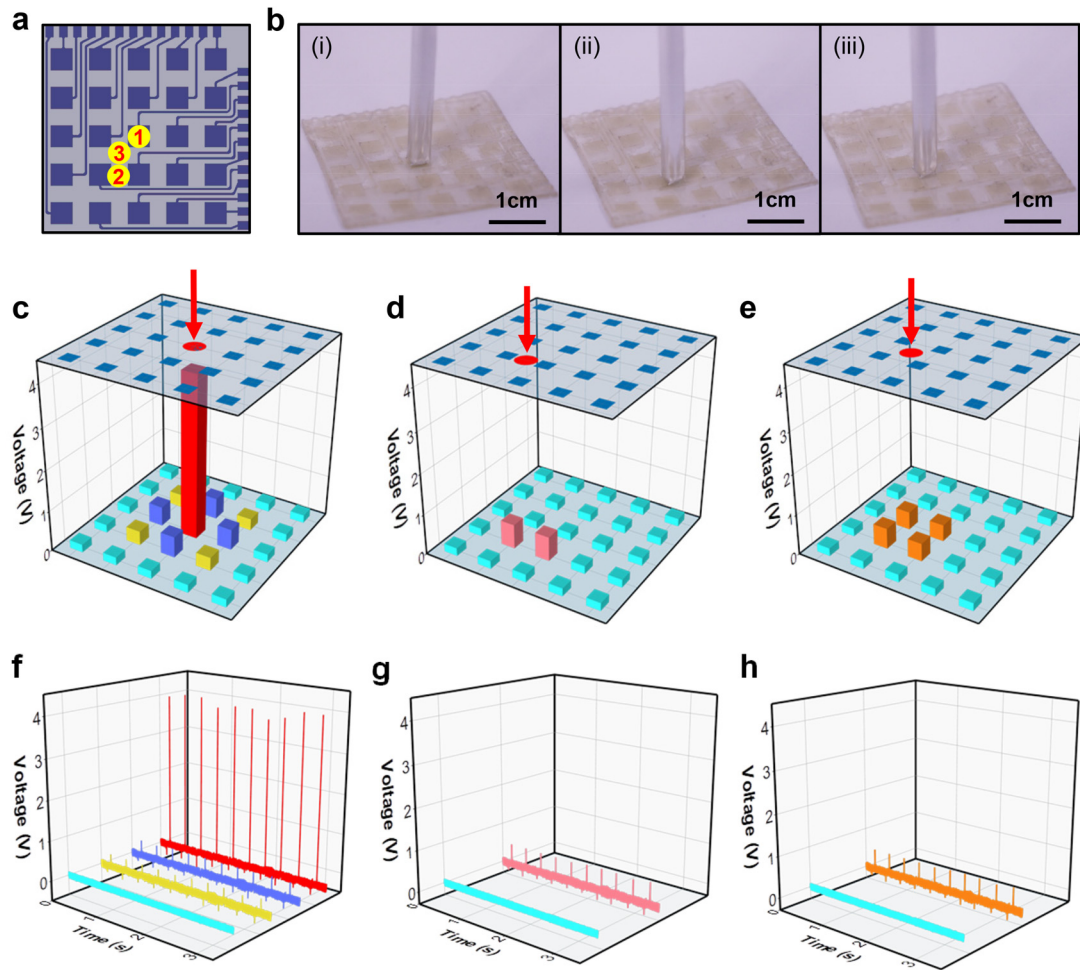


Fig. 3. Characterization of cross talk in a 5×5 sensor array. (a) Three loading positions, including sensing unit (1), interval (2) and intersection (3). (b) Optical images of the 5×5 sensor array under an aluminium bar with three loading positions, sensing unit (i), interval (ii) and intersection (iii). (c–e) Output voltages of each sensing units of the 5×5 sensor array with corresponding loading positions. (f–h) The corresponding voltage signals of sensing units under three loading positions. The voltage signals are classified according to their mean values in (c–e).

ing units, and the other is the interference of the cross talk. The voltage output as a function of the time of all sensing units can be found in Fig. 3g and Fig. S5e. Similarly, OC voltage output generated in four adjacent sensing units can be also found when the pressure was loaded in the middle of them (Fig. 3b–iii), yielding voltages of 0.5V, 0.45V, 0.45V, and 0.4V (Figs. 3e, h, and S5f), respectively. In general, it is obvious that when the load was applied on the sensing unit, negligible outputs were generated on 8 adjacent units, with the maximum output only taking 8.4% of the output generated in the loaded units. If the loading points were not directly on the sensing unit (interval or interaction of sensing units), slight outputs were generated in the adjacent sensing units with the maximum output taking 10.8% compared to that loaded on sensing units, indicating notable crosstalk suppression has been achieved in the sensor array.

Next, external loads with different widths, covering 1, 2, 3-unit widths with the weight of 3 g, 7.5 g, and 13 g were applied to the sensor array, as shown in Fig. 4a–c insets. Fig. 4a–c show the spatial distribution of 100 sensing units under corresponding loadings, where remarkable output voltages with the maximum amplitude at 0.06 V as well as a satisfied signal-to-noise ratio of 6 were generated in units under pressure while negligible output signals were generated in the adjacent sensing units, proving the excellent capability in crosstalk suppressing. Besides, the sensor array was loaded with the loading trace “2021” (Fig. 4d) and loading pattern “CITYU” (Fig. 4g) to further demonstrate its uniformity and sensitivity in response to external stimuli. Aluminum bar serving

as the loader was utilized to load on the sensing units (Fig. 4e) in the trace of “2021” with the pressure at 2.5 kPa, and the corresponding spatial distribution of output signals is shown in Fig. 4f, where the maximum output voltage of 0.3 V was generated in loaded sensing units while greatly suppressed the output signals in adjacent units. Besides, 3D printed loading objects “CITYU” with aluminum foil covered at the bottom (Fig. 4h) was applied to identify the sensing performance. Similarly, the maximum electric output of 0.3 V was generated in the loaded sensing units (Fig. 4i) while negligible output in adjacent units, proving not only the capability in crosstalk suppression but also excellent uniformity of the sensor array.

4. Conclusion

In summary, a soft, transparent, single electrode triboelectric sensor array with high pixel density (100 sensing units within the overall size of $7.5 \text{ cm} \times 7.5 \text{ cm}$) was developed through an easy and low-cost fabrication process. The single sensing unit shows good sensitivity at 0.11 V/kPa in a wide range of pressure from 10 to 65 kPa capable of distinguishing various pressing loads accurately, from gentle touching to hard pressing. Besides, the sensing unit showed excellent stability and durability with unchanged performance after more than 2000 cycles of pressure loading and 100 cycles of mechanical bending. In addition, owing to the difference in height and dielectric properties between the sensing unit and the connection part, both the crosstalk arising from

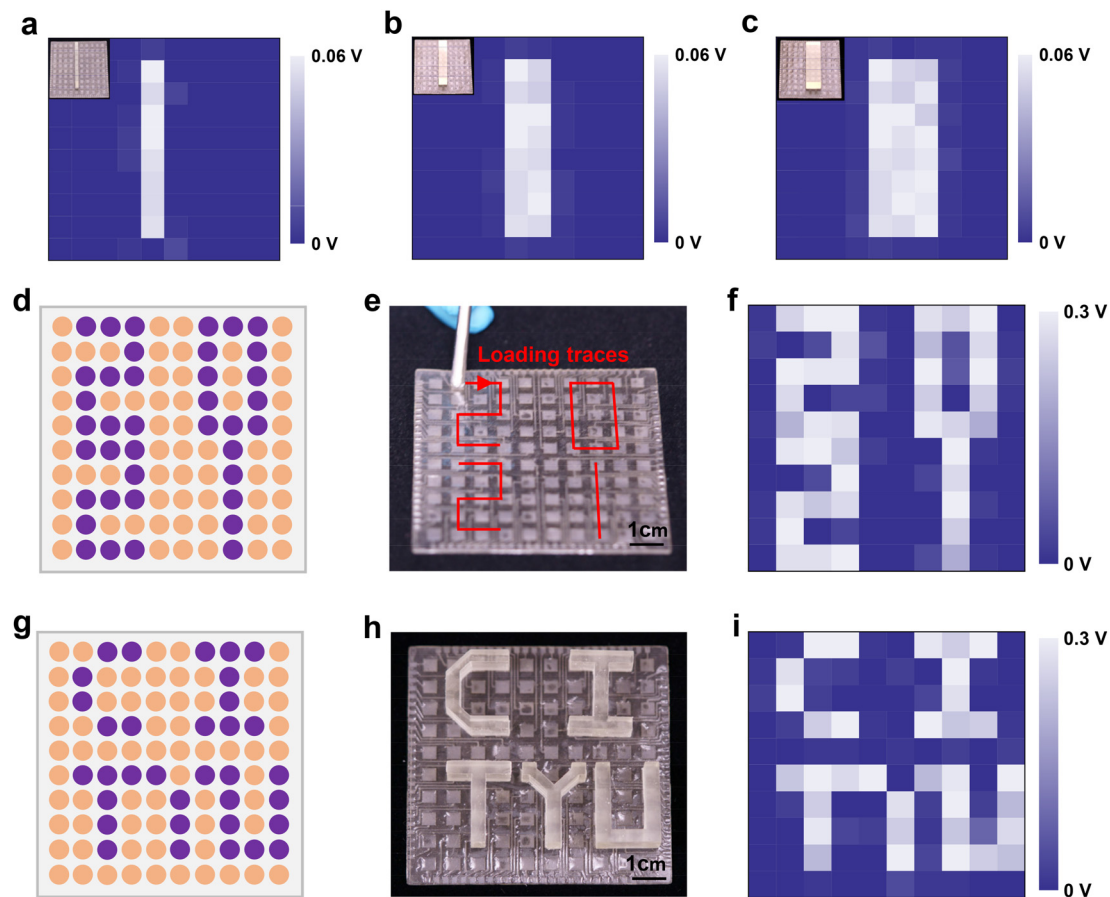


Fig. 4. Demonstrations of the 10×10 sensor array with different external loading manners. (a–c) Spatial voltage output of the sensor array under three types of width loader. (d) Schematic illustration of the loading trace “2021”. (e). Optical image of the sensor array loaded by aluminium bar in the trace of “2021”. (f) Corresponding voltage output of each sensing unit under the loading trace of “2021”. (g) Schematic illustration of the loading pattern “CITYU”. (h) Optical image of the sensor array loaded with the designed pattern “CITYU”. (i) Corresponding voltage output of each sensing unit under the loading pattern “CITYU”.

sensing units and electrodes can be greatly suppressed with the maximum crosstalk of only 10.8%. Demonstrations of loading trace in patterns of numbers “2021” and letters “CITYU” indicated the excellent uniformity and reproducibility of the sensor array, which presents its potential application in precise locating and human-machine interfaces.

Declaration of competing interest

The authors declare that they have no conflicts of interest in this work.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.fmre.2022.01.021.

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